Task 1 Report: Data Understanding and Related Work

Alzheimer's disease classification from structural MRI (OASIS-1) with a GCN + CNN + attention model

CSE 475 · Section 1 · Fall 2026 · Group: 11 · Track: GCN + CNN + Attention · Dataset: OASIS-1 (processed MRI, Kaggle)

1. Summary of findings

- **Data are clean and uniform.** 436 sessions from 416 subjects; every session has one image file and one CSV row (436 of 436 matched); all volumes are 91 × 109 × 91 voxels at 2 mm, orientation LAS, float32.
- **Severe class imbalance.** CDR 0 / 0.5 / 1 / 2 = 336 / 70 / 28 / 2 sessions (168:1). CDR 2 has only 2 subjects, so it cannot be learned or cross-validated on its own and is merged into CDR 1+.
- **Age is the dominant confound.** Nondemented subjects average 43.8 years and demented subjects 76 to 82. Age alone separates the classes in the full set (single-feature AUC 0.855). In subjects aged 60 and over the gap disappears (75.9 vs 76.8 years, $p = 0.44$) and the classes are balanced (98 vs 100 subjects), so the main experiment uses this subset.
- **A global brain-volume baseline is strong.** In the 60+ subset, normalized whole-brain volume (nWBV) alone gives a cross-validated AUC of about 0.70, and age + nWBV about 0.74. An image model has to beat this to show it learned more than global atrophy.
- **Intensity and quality.** Raw per-volume maximum varies from 1.18 to 2.97, so volumes need per-volume normalization. Global intensity does not separate the classes in the 60+ subset. About 7% of scans were flagged, including at least one clear skull-stripping or registration failure (OAS1_0351, CDR 2).
- **Research gap.** Six graph-based studies (2023 to 2026) build graphs from atlas regions, subject similarity or connectivity. None builds a graph from the pixels of each MRI image, none reports OASIS-only results, and leakage controls are often unstated.

2. Dataset

The data are the Kaggle dataset "PROCESSED MRI Scans for Alzheimer's Detection" (ninadaithal/oasis-1-shinohara), derived from the OASIS-1 cross-sectional collection (Marcus et al., 2007). OASIS-1 covers 416 right-handed adults aged 18 to 96, of whom 100 aged over 60 have very mild to mild Alzheimer's disease. All participants with dementia (CDR above 0) were diagnosed with probable Alzheimer's disease.

Table 1. Dataset summary.

Item	Value
Source	Kaggle "PROCESSED MRI Scans for Alzheimer's Detection" (ninadaithal/oasis-1-shinohara), derived from OASIS-1 (Marcus et al., 2007)
Subjects / sessions	416 subjects / 436 sessions (20 subjects have a second session, MR2)
Image files	436 NIfTI files (.nii), one per session; 3.61 MB each, about 1.57 GB in total
Image geometry	91 × 109 × 91 voxels, 2 × 2 × 2 mm, orientation LAS, float32; identical in all 436 files
Preprocessing (as described on the Kaggle page)	Skull-stripped, linear (12 degrees of freedom) registration to 2 mm MNI152, Shinohara intensity normalization
Labels	Clinical Dementia Rating (CDR) 0 / 0.5 / 1 / 2 = 336 / 70 / 28 / 2 sessions (77.1% / 16.1% / 6.4% / 0.5%)
Age and sex	18 to 96 years (mean 51.4); 268 female, 168 male; all right-handed

Item	Value
Other CSV columns	Educ (235 rows present), SES (216), MMSE (235), eTIV, nWBV, ASF (all 436), Delay (20)
Problem type	Supervised classification: binary (CDR 0 vs above 0) and three-class (CDR 0 / 0.5 / 1+). The input is a 3D T1 volume that we convert to 2D axial slices and then to per-image graphs

3. Data quality and structure

Linking images to labels. Session IDs were parsed from the file names and matched to the CSV: 436 matched, 0 images without a CSV row, 0 CSV rows without an image, and no duplicate IDs. Each session has exactly one file.

Repeat sessions. 416 rows are first sessions (MR1) and 20 are second sessions (MR2). All 20 repeat subjects are nondemented and under 60 (238 sessions but 218 subjects under 60). They are therefore absent from the 60+ subset, but any full-set experiment must keep both sessions of a person on the same side of a split.

Table 2. Missing values in the CSV.

Column	Missing	Percent	Reason
Delay	416	95.4%	Only the 20 repeat sessions carry a value
SES	220	50.5%	Collected mainly for older subjects
Educ	201	46.1%	Collected mainly for older subjects
MMSE	201	46.1%	Collected mainly for older subjects

Table 3. Fraction missing by age group (CDR is complete in every group).

Age group	MMSE	SES	Educ
Under 30	1.00	1.00	1.00
30 to 39	0.88	0.88	0.88
40 to 49	0.58	0.61	0.58
50 to 59	0.33	0.33	0.33
60 to 69	0.00	0.15	0.00
70 to 79	0.00	0.11	0.00
80 and over	0.00	0.04	0.00

Interpretation. Missingness is structural, not random: MMSE, education and SES are missing for every subject under 30 and for none aged 60 or over. Imputing them would invent data, and MMSE should not be a model input anyway because it is a near-proxy for the label. CDR is present for all 436 rows (the younger subjects are recorded as CDR 0).

4. Demographic and clinical distributions

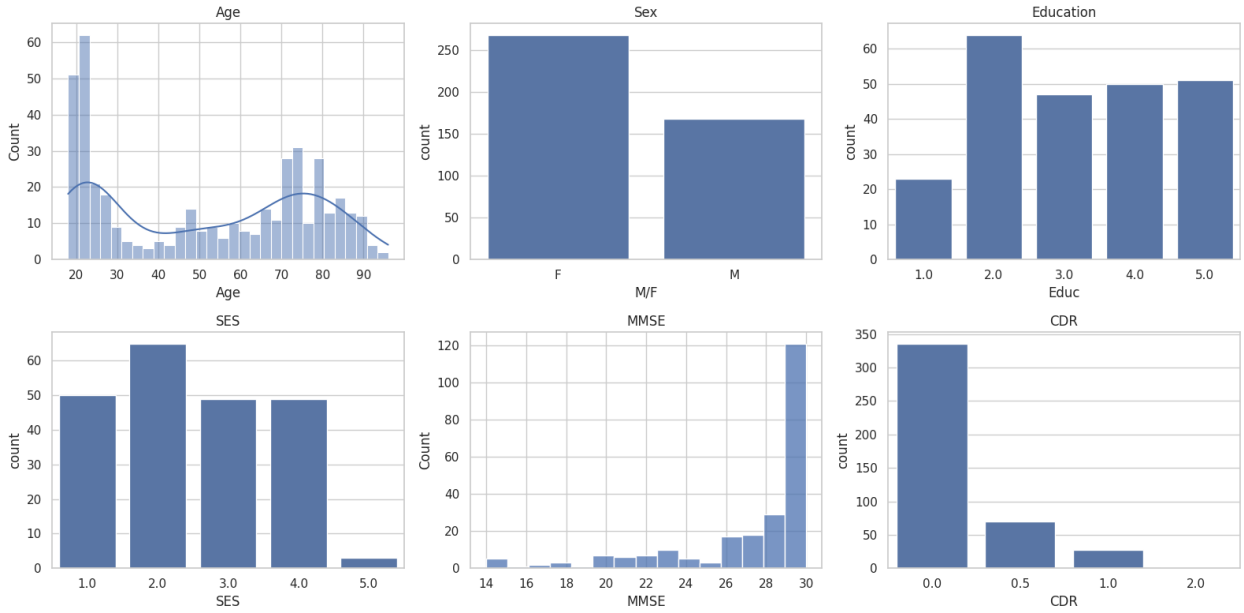


Figure 1. Distributions of age, sex, education, SES, MMSE and CDR (notebook section 5).

Interpretation. Age is bimodal: a young cluster in the early 20s (the tallest bin holds about 60 subjects) and a broad older group peaking around 70 to 80. There are more women (268, 61.5%) than men (168). Education and SES exist only for the older half of the sample. MMSE is concentrated near the ceiling of 30 with a long tail down to 14, and CDR is heavily skewed toward 0.

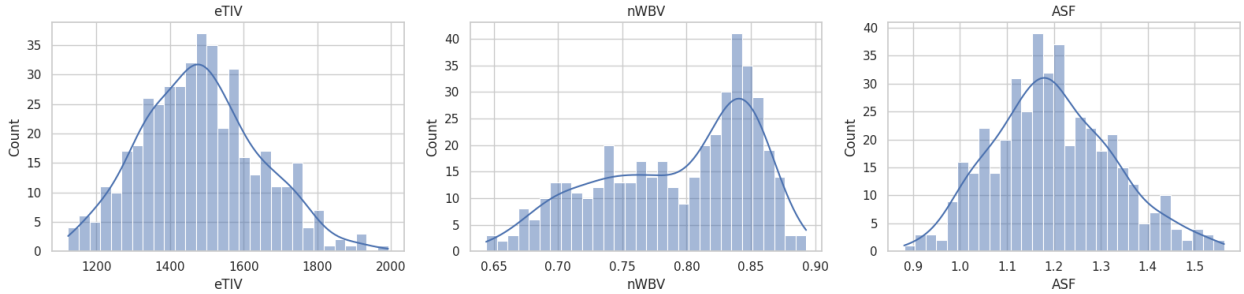


Figure 2. Distributions of eTIV, nWBV and ASF (notebook section 5).

Interpretation. Estimated total intracranial volume (eTIV) is roughly bell-shaped around 1,480 mm³ (range 1,123 to 1,992), and the atlas scaling factor (ASF) mirrors it. Normalized whole-brain volume (nWBV) is bimodal, with a peak near 0.84 from young adults and a long left tail toward 0.65 from older and demented subjects.

5. Class balance and the age confound

Table 4. Class balance over all 436 sessions (notebook section 12). Class weight = N / (k × n), for a weighted cross-entropy loss.

CDR class	Sessions	Percent	Ratio to majority	Class weight
Nondemented (0)	336	77.1	1.0	0.324
Very mild (0.5)	70	16.1	4.8	1.557
Mild (1)	28	6.4	12.0	3.893
Moderate (2)	2	0.5	168.0	54.500

Table 5. Subject-level label schemes (one row per subject).

Scheme	Classes (subjects)	Majority : minority
All subjects, binary	316 nondemented / 100 demented	3.16 : 1
Aged 60+, binary	98 nondemented / 100 demented	1.02 : 1
Aged 60+, three-class	CDR 0: 98 / CDR 0.5: 70 / CDR 1+: 30	3.27 : 1

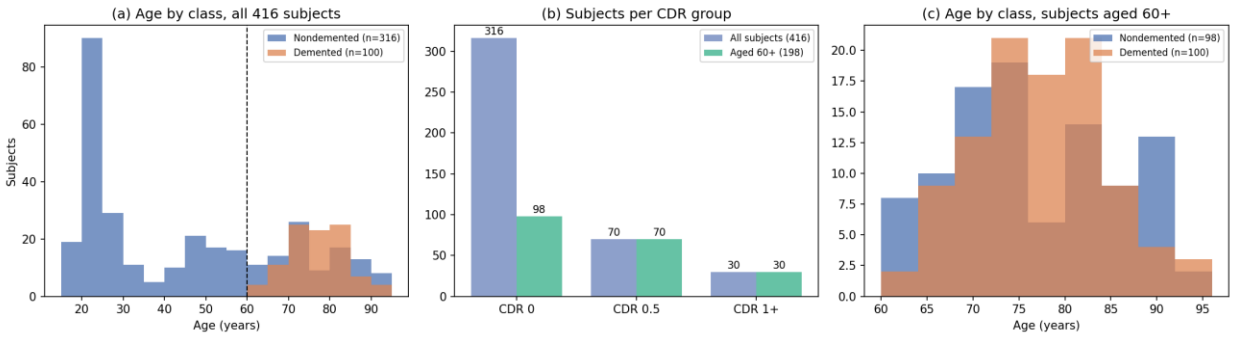


Figure 3. Age and class balance. (a) Age by class in all 416 subjects; the dashed line marks 60 years. (b) Subjects per CDR group, all subjects vs aged 60+. (c) Age by class in the 60+ subset. Generated from the OASIS CSV, first session per subject.

Interpretation. The session-level imbalance of 168:1 is driven entirely by CDR 2, which has two scans from two subjects. A class that small cannot be split across five folds and its weight of 54.5 would destabilize training, so CDR 2 is merged with CDR 1 (30 subjects). The binary imbalance of 3.2:1 in the full set is mostly an age effect: panel (a) shows that every demented subject is 62 or older while 218 of the 316 nondemented subjects are under 60. Restricting to subjects aged 60 and over removes the imbalance (98 vs 100) and leaves overlapping age distributions (panel c).

6. Clinical variables versus CDR

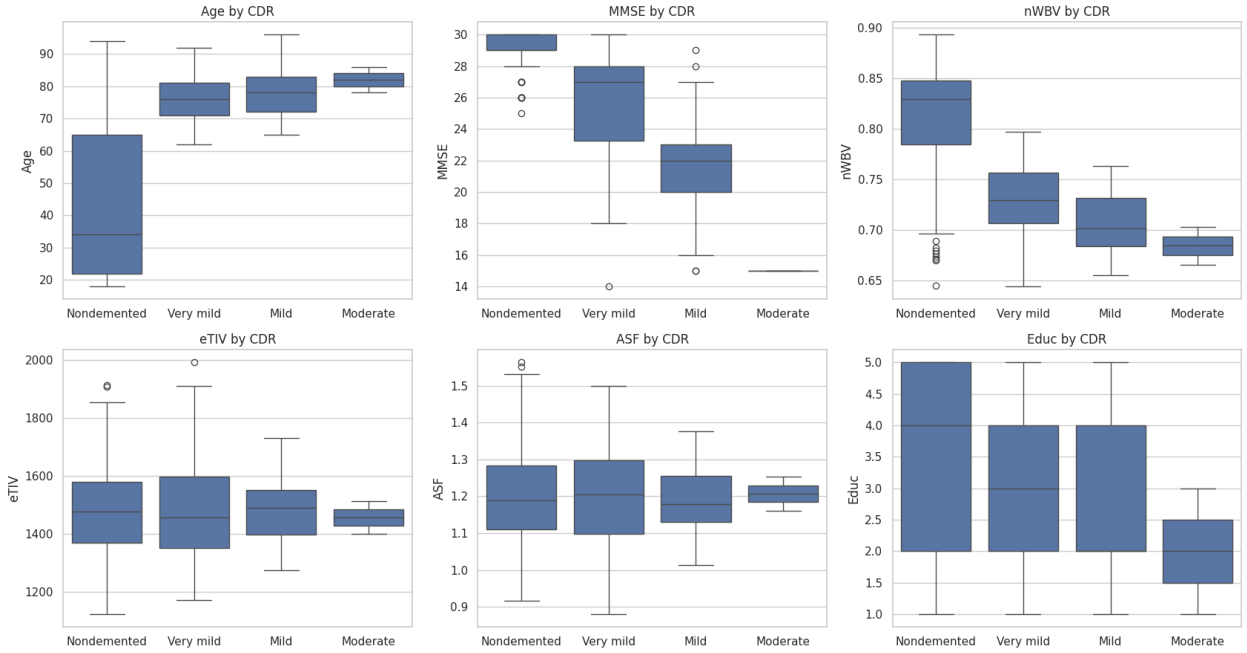


Figure 4. Age, MMSE, nWBV, eTIV, ASF and education by CDR group, labelled sessions (notebook section 7).

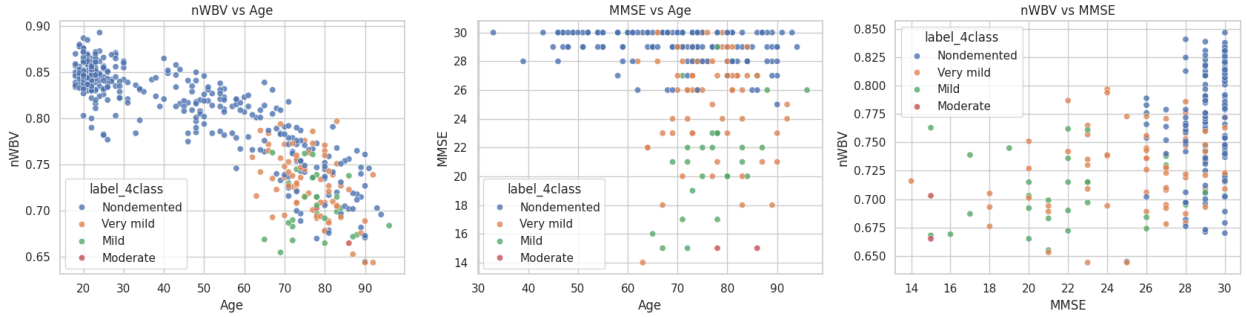


Figure 5. nWBV vs age, MMSE vs age, and nWBV vs MMSE, coloured by CDR group (notebook section 7).

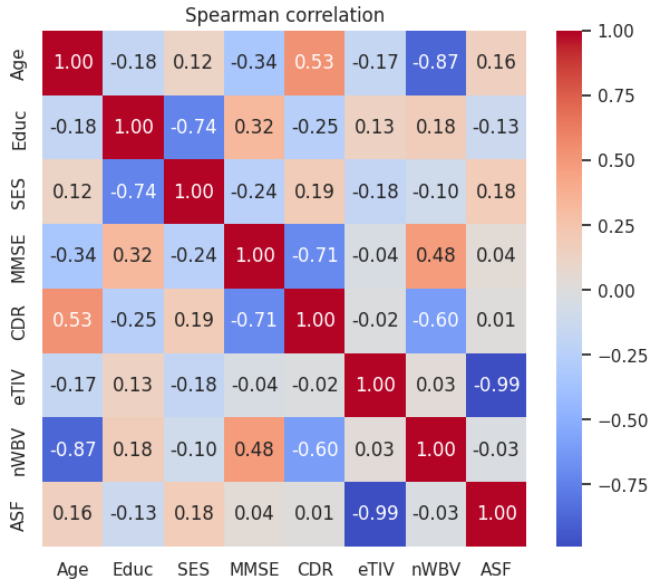


Figure 6. Spearman correlation matrix of the numeric columns (notebook section 8).

Table 6. Kruskal-Wallis tests across the four CDR groups, and a chi-square test of sex against the binary label (notebook section 9).

Variable	Kruskal-Wallis H	p-value
Age	122.37	2.4×10^{-26}
MMSE	117.68	2.4×10^{-25}
nWBV	159.19	2.7×10^{-34}
eTIV	0.20	0.977
Education	15.38	0.0015
Sex vs demented (chi-square, 1 dof)	0.21	0.645

Interpretation. Age rises steeply with CDR and nondemented subjects span the whole adult range (18 to 94), whereas the demented groups sit between 62 and 96. MMSE and nWBV fall with CDR, but both overlap heavily between neighbouring groups, especially nondemented and very mild. eTIV and ASF do not differ between groups ($p = 0.98$), which is expected for head-size measures. Education differs slightly ($p = 0.0015$), possibly because the older cohort had less schooling. The correlation matrix shows nWBV with age at −0.87, MMSE with CDR at −0.71, nWBV with CDR at −0.60 and age with CDR at 0.53. eTIV and ASF correlate at −0.99 and carry the same information, so only one should ever be used in a tabular model. In the scatter plot, nWBV declines with age even in nondemented subjects, and demented subjects sit lower than nondemented subjects of the same age only with considerable overlap.

6.1 What changes when we restrict to subjects aged 60 and over

Table 7. Subjects aged 60+ (one row per subject): nondemented vs demented. Mann-Whitney tests except where noted. Computed from the OASIS CSV.

Variable	Nondemented (n = 98)	Demented (n = 100)	p-value
Age (years)	75.9 ± 9.0	76.8 ± 7.1	0.44
nWBV	0.75 ± 0.04	0.72 ± 0.04	2.8×10^{-7}
eTIV (mm^3)	1,439 ± 150	1,484 ± 168	0.059

Variable	Nondemented (n = 98)	Demented (n = 100)	p-value
Education code	3.27 ± 1.31	2.82 ± 1.30	0.016
MMSE (not a model input)	29.0 ± 1.2	24.3 ± 4.2	4×10^{-20}
Sex, female / male	72 / 26	59 / 41	0.045 (chi-square)

Table 8. Simple tabular baselines on the 60+ subset (198 subjects): logistic regression, 5-fold subject-grouped stratified cross-validation, AUC mean ± sd. Computed from the OASIS CSV.

Features	AUC
Age only	0.45 ± 0.11
nWBV only	0.70 ± 0.12
Age + nWBV	0.74 ± 0.11
Age + nWBV + eTIV + sex	0.71 ± 0.10

Interpretation. In the full set, age alone gives a single-feature AUC of 0.855, so a classifier could score well by recognising old brains. In the 60+ subset the age gap is gone ($p = 0.44$; age-only AUC 0.53 overall and 0.45 in cross-validation, which is chance level for 198 subjects), while nWBV still separates the groups ($p = 2.8 \times 10^{-7}$). Sex is mildly imbalanced (women are 73% of nondemented and 59% of demented subjects), so we will report results by sex as a check. The practical consequence is a bar for Task 2: an image model should beat roughly 0.74 AUC, the age + nWBV baseline, to show it learned more than global brain volume. nWBV is itself computed from the scans, so these baselines are for comparison only and nWBV is never a model input.

7. Image inventory and visual inspection

Table 9. Image inventory (notebook sections 10, 13 and 14).

Property	Result
Files	436 NIfTI files; 434 follow the pattern "<ID>_mpr_n#_anon_sbj_111_normalised.nii" and 2 follow "<ID>_mpr_n#_fdcanon_sbj_#.#dfp_normalised.nii"
Shape	(91, 109, 91) in all 436 files
Voxel size	2.0 × 2.0 × 2.0 mm in all 436 files; field of view 182 × 218 × 182 mm
Orientation / data type	LAS in all files / float32; single channel; no NaN or infinite voxels in the 150 scans measured
Resolution vs label	No confound: all classes share one shape and voxel size

The two files with a different name pattern should be identified and checked to confirm they went through the same preprocessing as the other 434.

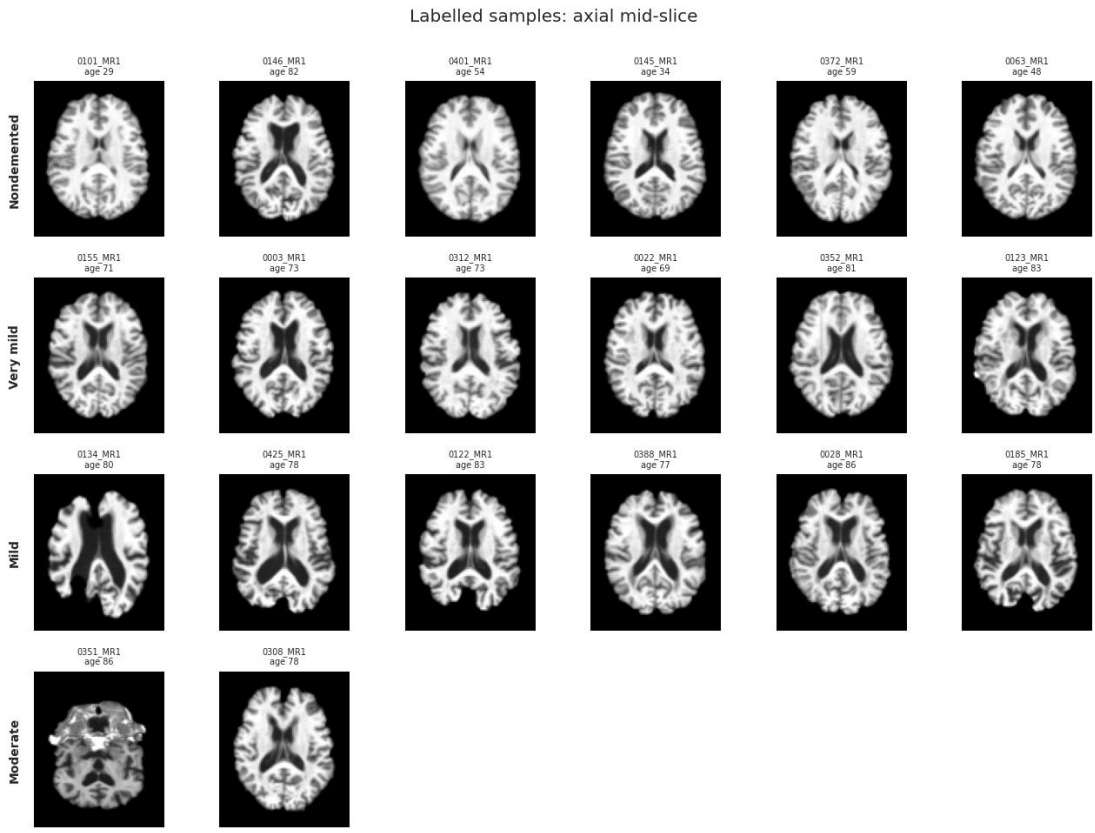


Figure 7. Labelled sample grid: axial mid-slice of six random subjects per class (two for CDR 2), scaled to the 1st to 99th percentile (notebook section 15).

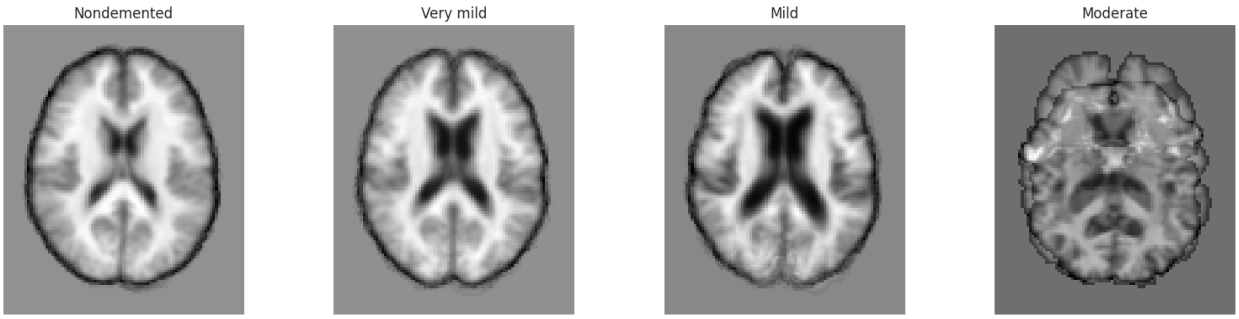


Figure 8. Group-mean axial slice per CDR class: 15 scans per class, 2 for CDR 2 (notebook section 18).

Observations.

- Ventricles are larger and sulci wider in the very mild and mild groups than in young nondemented brains, and the group means show ventricular enlargement from nondemented to very mild to mild.
- CDR 0 and CDR 0.5 look similar, and older nondemented subjects also show large ventricles (for example OAS1_0146, age 82). Age therefore produces visible atrophy by itself, which is the age confound again. CDR 0 vs 0.5 is likely the hardest pair to separate.
- OAS1_0351 (CDR 2, age 86) shows non-brain tissue and a lower anatomical level at the mid-slice, which points to a skull-stripping or registration failure. With only two CDR 2 scans, the Moderate group mean is dominated by this one image and is not interpretable.
- OAS1_0134 (CDR 1, age 80) has very large, distorted ventricles. This may be severe real atrophy or an artifact and should be inspected. OAS1_0123 (CDR 0.5) has a bright spot at the left edge.

- The class-similarity analysis in the notebook (correlation of class-mean slices after removing the grand mean) gives Very mild – Mild as the most similar pair (0.40) and Very mild – Moderate as the most distinct (−0.82). The second figure reflects the failed Moderate image rather than biology, and the method (one mid-slice, 25 scans per class) is crude.

8. Intensity and image quality

Table 10. Per-scan statistics on a random sample of 150 labelled scans (notebook section 16).

Statistic	Mean ± sd	Range
Raw maximum intensity	2.23 ± 0.29	1.18 to 2.86
99th percentile (p99)	1.083 ± 0.046	1.05 to 1.54
Brain fraction (voxels above 10% of p99)	0.264 ± 0.009	0.214 to 0.292
Mean brain intensity	0.705 ± 0.012	0.668 to 0.761
SNR (brain mean / background sd)	87.7 ± 4.5	53.1 to 104.2
Contrast	0.643 ± 0.040	0.576 to 0.747
Sharpness (variance of Laplacian, mid-slice)	0.058 ± 0.020	0.030 to 0.198

Table 11. Mean quality metrics by class (150-scan sample; the Moderate class has 2 scans).

Class	Sharpness	SNR	Contrast	Brain fraction
Nondemented	0.052	87.9	0.630	0.266
Very mild	0.074	88.3	0.682	0.257
Mild	0.080	83.8	0.698	0.262
Moderate (n = 2)	0.128	84.2	0.705	0.237

Normalization. The raw maximum varies by a factor of 2.4 between scans, but the 99th percentile is stable (mean 1.083). The maximum is therefore set by a handful of bright voxels and is a poor scaling reference. We will normalize each volume with its p99 or by z-scoring brain voxels, never by the per-image maximum. No NaN voxels were found and saturation is about zero.

Quality metrics versus class. Sharpness and contrast rise with severity, which is what more cerebrospinal-fluid edges in atrophic brains would produce, so these differences should not be read as image-quality problems. They do mean that simple image statistics can carry some label information, which is why we check for shortcuts below.

Shortcut checks on the 60+ subset (198 subjects). The per-volume maximum gives an AUC of 0.488 and the mean brain intensity 0.431 with a bootstrap 95% confidence interval of 0.349 to 0.512, which includes 0.5. The number of scans averaged into each volume (n3 / n4 / n6 in the file name) is also unrelated to the class (n3: 6 nondemented, 13 demented; n4: 91, 87; n6: 1, 0; Fisher exact p = 0.147). We found no reliable global-intensity shortcut.

Slice profile. The median number of brain voxels per axial slice, across the 436 volumes, rises from zero at the bottom to a plateau of about 5,400 voxels around slices 30 to 45, and is zero from about slice 78. Slices 79 to 90 are empty. We will keep the slices where the median brain area is at least 50% of the peak (about slices 16 to 65 on the observed curve; the exact range comes from the threshold rule) and take every third slice, giving about 17 slices per subject.

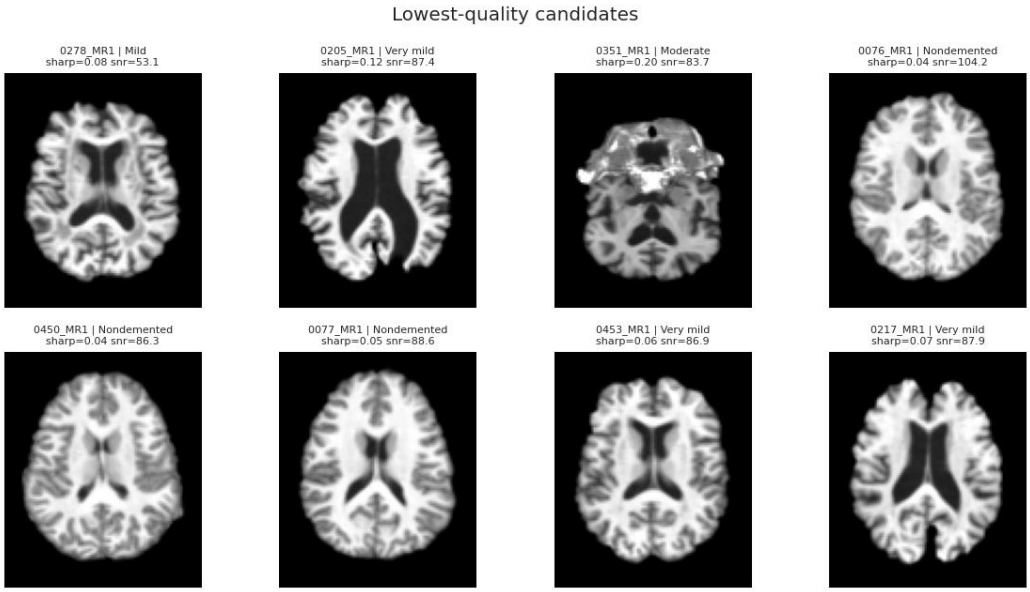


Figure 9. The eight lowest-quality candidates by robust z-score of sharpness, SNR, contrast, brain fraction and saturation (notebook section 16).

Flagged scans. 10 of the 150 sampled scans (6.7%) were flagged on at least one metric (robust z-score above 3.5). Across all 436 volumes, a robust-z check on per-volume statistics flagged OAS1_0199 and OAS1_0316 (maximum intensity) and OAS1_0016, 0085, 0123, 0205 and 0278 (mean brain intensity). OAS1_0351 (Moderate) shows non-brain tissue, and OAS1_0278 (Mild) has the lowest SNR (53). None are removed at this stage. All their slices will be inspected, and any exclusion will be documented with the reason.

9. Modelling readiness and design decisions

The notebook's example of a subject-grouped stratified 5-fold split on all 436 sessions gives test folds of 85 to 89 sessions with 17% to 34% demented, which shows how unstable folds are when only 100 subjects are demented. This is another reason for the more balanced 60+ subset.

Table 12. Decisions for Task 2 and the evidence behind them.

Decision	Choice	Evidence
Main population	Subjects aged 60+ (198: 98 CDR 0, 100 CDR above 0)	Age gap disappears (p = 0.44); balance 1.02:1
Labels	Binary CDR 0 vs above 0 (main); three-class 0 / 0.5 / 1+ (extension); CDR 2 merged with CDR 1	CDR 2 has 2 subjects
Split	Subject-wise stratified 5-fold CV; graph built after the split	Repeat sessions; leakage rule; unstable folds
Sessions	First session (MR1) only	All MR2 rows are under 60
Image input	Fixed axial slice range (about 16 to 65, every third slice); subject ID kept on every slice	Slice profile
Intensity	Per-volume normalization (p99 or z-score of brain voxels)	Raw maximum varies 1.18 to 2.97
Disallowed inputs	CDR and MMSE (label proxies); nWBV, eTIV, ASF (derived from the scans; baselines and EDA only)	Avoid label leakage
Baselines	Age + nWBV logistic regression (AUC about 0.74) plus CNN baselines	Bar for the image model

Decision	Choice	Evidence
Metrics	AUC, sensitivity, specificity, macro-F1, per-class recall, confusion matrix; mean ± sd across folds	Imbalance; course rubric
Quality control	Inspect flagged scans; document exclusions (for example OAS1_0351)	Registration failure seen
Sex	Report results by sex (60+: 72/26 vs 59/41 female/male)	Mild imbalance, p = 0.045

10. Related work

Six studies from 2023 to 2026 are compared. Each builds a graph for MRI-based Alzheimer's disease (AD) classification or prognosis and uses a GCN, GraphSAGE or GAT. Every entry was checked against the text of the paper. Tables 1 and 2 describe each study, Table 3 gives the graph construction, GCN type and attention type, Table 4 gives strengths and limitations, and Table 5 gives the gap each study leaves and its relation to our work. Rows use the same number in every table.

Table 13. Related work: Studies.

#	Paper	Dataset	Task
1	Qin, Liu, Lu, Zhu (2023). Research on classification method of multimodal magnetic resonance images of Alzheimer's disease based on generalized convolutional neural networks. J Biomed Eng 40(2):217–225	ADNI; 91 subjects (44 AD, 47 NC); T1 sMRI + resting-state fMRI	AD vs NC
2	Yang, Li, Gan, Huang, Shi, Zhou (2024). An Alzheimer's Disease classification network based on MRI utilizing diffusion maps for multi-scale feature fusion in graph convolution. Math Biosci Eng 21(1):1554–1572 (online Dec 2023)	ADNI, 1.5T T1; 1,292 samples (338 AD, 422 MCI, 532 CN); 116 AAL region features per sample	AD vs CN, AD vs MCI, CN vs MCI, 3-class
3	Qu, Yao, Liu, Wang (2023). A graph convolutional network based on univariate neurodegeneration biomarker for Alzheimer's disease diagnosis. IEEE J Transl Eng Health Med 11:405–416	ADNI, Aβ+ subjects; 318 longitudinal subjects for training (84 AD, 102 MCI, 132 CU) and 128 for validation (32, 46, 50); FreeSurfer cortical thickness, left hemisphere, 34 regions	AD vs CU, AD vs MCI
4	Song et al. (2024). s²MRI-ADNet: an interpretable deep learning framework integrating Euclidean-graph representations of Alzheimer's disease solely from structural MRI. Magn Reson Mater Phy 37:845–857	ADNI, AIBL, EDSD, OASIS; 3,377 subjects; sMRI only (gray matter maps, 91×109×91 at 2 mm)	NC vs AD; 10-fold CV within ADNI and cross-database testing
5	Li, Feng, Wang, Hou, Wan, Zhang, Hu (2026). Graph-based multimodal fusion framework with correlation-aware learning for Alzheimer's disease prediction. Big Data Min Anal 9(3):687–704	ADNI; 417 subjects (103 CN, 103 EMCI, 105 LMCI, 106 AD); T1 sMRI + 54 SNPs near APOE	CN vs AD, CN vs MCI, MCI vs AD; 5-fold CV
6	Ashimgaliyev, Zhumadillayeva, Mussabek, Qin (2026). Longitudinal graph neural networks for prognostic modeling of Alzheimer's disease progression. 2026 Int Conf Adv AI and ML (AAIML), pp. 610–619	ADNI; about 600 MCI subjects; longitudinal MRI, FDG-PET and amyloid PET	MCI-to-AD conversion within 5 years

Table 14. Related work: Method and headline results.

#	Method	Headline results
1	3D hybrid-attention residual U-Net for sMRI plus a U-shaped GCN (Graph U-Net style) for fMRI; features fused, binary PSO feature selection, then a machine-learning classifier	Fusion: ACC 88.89%, SEN 100%, SPE 77.78%, G-mean 88.19%. U-GCN alone: ACC 83.33%. One random 7:1:2 split (about 18 test subjects)

#	Method	Headline results
2	Population graph (nodes are patients); diffusion maps at several neighbour counts give multi-scale node features; three GCNs, self-attention over the scales, metric-learned adjacency	Accuracy only: AD-CN 0.887, AD-MCI 0.705, CN-MCI 0.825, 3-class 0.671 (average over random 7:1:2 splits)
3	GCN on a population graph; node features are 34 univariate neurodegeneration biomarkers (UNB); feature attention module	AD vs CU: test ACC 94.25%, SEN 98.39%, SPE 89.13%, AUC 95.30%; validation ACC 93.90%. AD vs MCI: validation ACC 82.05%
4	Two channels: gray matter map through CNN plus Transformer; regional radiomics similarity network through GraphSAGE plus MLP; features concatenated	ADNI 10-fold: ACC 92.1%, SEN 88.0%, SPE 94.2%, AUC 0.962. Cross-database: ACC 91.4%, SEN 77.2%, SPE 96.4%, AUC 0.938
5	Brain-region graph (sMRI) and gene-ontology graph (SNP); GCN with pooling; cross-attention; correlation-aware learning loss	CN vs AD: ACC 0.823 ± 0.034, AUC 0.822. CN vs MCI: ACC 0.773. MCI vs AD: ACC 0.747
6	Per-region 3D CNN embeddings as node features; spatiotemporal graph of 82 regions over visits; GAT plus GRU with temporal attention	AUC 0.89 (95% CI 0.85–0.93), C-index 0.84, BAC 86%, SEN 85%, SPE 87%. Best baseline (CNN-LSTM): AUC 0.84

Table 15. Related work: Graph construction, GCN type and attention type.

#	Graph construction	GCN type	Attention type
1	Nodes: atlas brain regions. Edges: correlation between region time series (functional connectivity). Extra node features appended. One graph per subject	Kipf-style GCN inside a Graph U-Net (graph pooling and unpooling, skip connections)	Hybrid (channel and spatial) in the 3D CNN branch; none in the GCN
2	Nodes: patients. Edges: Gaussian-kernel similarity of 116-dimensional region vectors, thresholded; a second, learned adjacency from metric learning; diffusion maps at several neighbour counts	Two-layer GCN, three copies (one per scale)	Self-attention over the multi-scale features
3	Nodes: patients. Edges: sum of phenotype similarity (age, gender, APOE; threshold on numeric differences). Node features: 34 UNB values	Spectral GCN (Kipf and Welling), 3 hidden layers	Feature attention on node features (softmax over a learned map)
4	Nodes: 90 AAL regions per subject. Node features: 25 radiomic features per region. Edges: Pearson correlation between regions, kept above 0.8	GraphSAGE (two residual blocks); MLP for edge strengths	Multi-head self-attention (Transformer) in the gray matter channel
5	Nodes: AAL3 regions; edges: Pearson correlation computed across samples, thresholded; second graph for the SNP gene-ontology hierarchy	Kipf GCN with global pooling (MRI) and hierarchical pooling (SNP); GAT as baseline	Cross-attention between modalities
6	Nodes: 82 regions at each visit. Spatial edges: anatomical adjacency or baseline correlation above 0.8. Temporal edges: same region at consecutive visits. Node features: 128-dimensional 3D CNN embedding plus 4 regional measures	GAT (8 heads, 2 layers) with GRU updates; not a GCN	Graph attention (spatial) plus temporal self-attention

Table 16. Related work: Strengths and limitations.

#	Strengths	Limitations
1	Graph model on fMRI beats the 3D CNN on the same subjects (83.3% vs 77.8%); compared against GraphSAGE, TAGCN and a higher-order GNN; fusion adds a further gain	Only 91 subjects; test set of about 18, so one subject equals 5.56 accuracy points (the headline 5.56% gain is one subject); one random split, no cross-validation or significance test; needs paired fMRI; no explainability analysis

#	Strengths	Limitations
2	Reduces dependence on a fixed adjacency matrix; ablation of diffusion maps, metric learning and fusion; several GNN baselines	Accuracy only (no AUC, sensitivity or specificity); random 7:1:2 splits, no cross-validation or confidence intervals; counts are samples, not subjects, so repeat scans of one person could fall in different splits (not stated); needs pre-extracted region vectors
3	Attention gives region-importance maps; compared with SVM, RF, LR, XGBoost, CNN, GAT and GCN; source code public	Small, ADNI-only cohort; left hemisphere only; the UNB features use a group atrophy pattern built from labelled AD and CU subjects, and whether test subjects were excluded from it is not stated; the graph includes all subjects as nodes
4	Large multi-site sample; cross-database testing; ablation of each channel; interpretability analysis; code public; same 91×109×91, 2 mm grid as our data	How splits were grouped by subject is not described; no significance test; OASIS is named OASIS in Table 1 but OASIS3 in the text (version unclear) and no OASIS-only result is reported; needs atlas and radiomics extraction
5	Explicit inter-modality correlation module; ablation over loss terms and modalities; 5-fold CV with mean ± std; biomarker and SNP interpretation	417 subjects, ADNI only, no external validation (stated by the authors); the region correlation is computed over all samples and it is unclear whether training folds only were used; text (0.812 ± 0.045) and Table 2 (0.823 ± 0.034) disagree for CN vs AD; needs SNP data
6	Subject-level split with all visits kept together; 95% bootstrap confidence intervals; DeLong tests; ablation of edges, modalities and CNN; integrated-gradient interpretation	Needs longitudinal MRI and PET (not available for us); ADNI only; ablation numbers differ between sections (removing temporal edges: AUC 0.82 in one section, about a five percent drop in another); the conclusion names the posterior cingulate while the results name hippocampus and entorhinal cortex

Table 17. Related work: Research gap and relation to our work.

#	Gap the study leaves	Relation to our work
1	The graph comes from fMRI time series, not from the image itself; very small sample	Shows an attention-CNN branch and a graph network can be combined; our graph comes from the structural image
2	Graph over patients built on pre-extracted region vectors, so images never become graphs; accuracy only	GCN with self-attention fusion on MRI, the nearest to our GCN + attention; we build a graph inside each image instead
3	Patient-level graph; age is used as an edge but not controlled	Attention on node features and phenotype edges; we handle age by using subjects aged 60 and over and splitting by subject
4	Atlas-based graphs, not image-derived; its OASIS cohort differs in age (NC 67.1 vs AD 75.0 years, $p < 0.001$) and no age control is described	Closest data match (OASIS, 91×109×91 at 2 mm) and our strongest comparison candidate, but it reports no OASIS-only result
5	Needs genetics; region graph computed across samples; no OASIS data	Reference design for cross-attention and for ablation plus 5-fold reporting like our Task 3
6	Needs longitudinal PET; uses region patches, not a graph over a whole slice	CNN embeddings as graph nodes is the idea behind our CNN + GCN; its split, confidence intervals and significance tests are the protocol we must follow

10.1 Why each study used a graph, and the gap it leaves

Table 18. Rationale for the model choice in each study.

#	Why this model	Gap it leaves
1	A 3D CNN suits the sMRI grid, but an fMRI connectivity network is irregular, so a CNN cannot be applied and a GCN is used	The graph comes from fMRI time series, not from the image; only 91 subjects
2	A fixed adjacency matrix is a weakness, so the adjacency is learned (metric learning) and diffusion maps give multi-scale features	Works on pre-extracted region vectors; the image never becomes a graph
3	A CNN cannot use correlations between individuals, so patients become nodes in a GCN with age, sex and APOE edges	Hand-engineered features; age is an edge, not controlled

#	Why this model	Gap it leaves
4	CNNs miss long-range and non-grid structure, so a Transformer covers long range and a GNN covers the region network	Graph comes from a fixed atlas plus radiomics
5	Concatenating modalities ignores their relationships, so each modality gets a graph and cross-attention links them	Needs SNP data; correlation graph computed across all samples
6	A single snapshot misses change, so a graph across visits models progression	Needs longitudinal PET and MRI

Why a GCN for our data. Alzheimer's atrophy is spread over several non-adjacent places: ventricles widen, sulci widen and cortex thins. A CNN relates distant regions only through deep stacks or large kernels. A graph over image regions lets information travel between distant regions in one or two hops, and attention can weight which regions matter. A Transformer over patches does something similar but needs more data; with about 198 subjects the sparse structure of a graph may be a lighter inductive bias. This is a hypothesis to test, not an established fact. The GCN is kept shallow (2 to 3 layers) to limit over-smoothing.

What we expect to learn. Because nWBV correlates with age at −0.87 and with CDR at −0.60, a model could be learning only global atrophy plus age. The informative result is whether the regional graph adds anything beyond a CNN-only model and beyond the age + nWBV baseline under subject-wise cross-validation. If node-importance maps point at ventricles, medial temporal regions and sulci, that supports biological validity; if they point at skull edges or non-brain tissue (as in OAS1_0351), the model is using a shortcut. If the GCN is no better than the CNN, that is still a valid finding.

Novelty, stated narrowly. Among the six studies reviewed, none builds a per-image superpixel or patch graph from T1 slices. Our contribution is that design (CNN features, a per-image graph and attention) together with a stricter protocol on OASIS-1: subject-wise splits, an age-controlled 60+ subset, 5-fold cross-validation, a significance test, an ablation of the graph itself and checked explanations. We do not claim that graph-from-image methods are new in general; superpixel graph networks exist in other imaging areas, and only these six papers were reviewed.

10.2 Research gap

Across the six studies: (1) none builds a graph from the pixels of each MRI image (superpixels or patches); graphs come from atlas regions, subject similarity or connectivity. (2) Five of six use ADNI only; the one study that includes OASIS reports pooled results, and its OASIS cohort shows a large age gap between normal and AD groups with no age control described. (3) Leakage controls are often unstated (subject-level splitting, graph built before splitting), and several headline gains rest on single random splits of small samples. (4) Ablations of the graph itself (construction method, edge weights) are rare.

Our proposed contribution: a GCN + CNN + attention model that builds a per-image graph (superpixel or patch nodes) from T1 slices, evaluated on OASIS-1 with subject-wise splits, an age-controlled subset of subjects aged 60 and over, 5-fold cross-validation, an ablation of the graph and a significance test against the best CNN baseline.

10.3 Also reviewed

- SinhaRoy and Sen (2024), GAN + CNN on 2D slices, 99.7% accuracy: not a graph method; no subject-level split described. Its class counts (3,200 / 2,240 / 896 / 64) match those of the Kaggle four-class set used by Lanjewar et al., although the paper states ADNI.
- Lanjewar, Parab and Shaikh (2023), CNN + KNN on the Kaggle four-class set, 99.58%: not a graph method (summarised from the group comparison table; not independently checked).
- Goyal, Rani and Singh (2024), AlexNet + LSTM with GAN augmentation on 2D slices: not a graph method (summarised from the group comparison table; not independently checked).
- Ahmad, Tarannum and Uddin (2026), survey of ML and DL for AD diagnosis, and Ali, Piana, Pardini and Garbarino (2025), review of GNN-based AD diagnosis (73 studies): background citations with no experiments of their own (summarised from the group comparison table; not independently checked).

Each figure is quoted from its paper under that paper's own cohort and split. The results are not comparable with each other or with our OASIS results. Studies 1 and 2 use single random splits on small samples, so their headline gains should be read with caution.

11. Planned approach (preview of Task 2 and Task 3)

- **Baselines:** a simple CNN, ResNet50, DenseNet121 and EfficientNet-B0 on the selected axial slices, plus the age + nWBV logistic-regression baseline.
- **Proposed model:** CNN features on superpixel or patch nodes of each slice, a per-image graph with edges from spatial adjacency or feature similarity, a shallow GCN with attention, and a classifier.
- **Ablation:** CNN only, CNN + attention, CNN + GCN, full model; edge weights on or off; superpixel vs patch graph; GCN vs GAT vs GraphSAGE.
- **Evaluation:** subject-wise 5-fold cross-validation, mean ± sd, a paired significance test against the best baseline, and explanations with Grad-CAM, LIME and GNNExplainer on a correct and a wrong prediction.

12. Limitations of this analysis

- Image statistics and quality flags come from a random sample of 150 of 436 scans, and the quality metrics are heuristic.
- The class-similarity and group-mean analyses use a single mid-slice, and the CDR 2 class has two scans, so conclusions about it are not reliable.
- We measured shape and voxel size but not registration quality; the preprocessing description comes from the Kaggle page.
- OASIS-1 is a single-site, cross-sectional collection with clinical (CDR-based) labels. Restricting to subjects aged 60+ leaves 198 subjects and limits generalization to younger adults.
- Numbers quoted from related work come from each paper's own cohort and split and are not comparable with each other or with our results.

Appendix A. Screening of the 11 candidate papers

Initial group screening table with the final decision for each paper. Titles marked “Title to confirm” in the screening table have been resolved from the papers. Rows 4, 8, 9, 10 and 11 are summarized from the screening table and were not independently checked against the full text.

#	Paper	Dataset	Method	Headline metrics	Decision
1	Research on classification method of multimodal magnetic resonance images of Alzheimer’s disease based on generalized convolutional neural networks. Qin et al., 2023, J Biomed Eng	ADNI; 91 subjects (44 AD, 47 NC); sMRI + rs-fMRI	gCNN: 3D HA-ResUNet (sMRI) + U-GCN (fMRI brain network); feature fusion, binary PSO feature selection, ML classifier	ACC 88.89%, SEN 100%, SPE 77.78%, G-mean 88.19% (best single modality, U-GCN: ACC 83.33%)	Included
2	s²MRI-ADNet: an interpretable deep learning framework integrating Euclidean-graph representations of Alzheimer’s disease solely from structural MRI. Song et al., 2024, MAGMA	ADNI, AIBL, EDSD, OASIS; 3,377 subjects; sMRI only	Dual channel: GMV maps with CNN-Transformer + regional radiomics similarity network (R2SN) with GraphSAGE and MLP	Intra-database ACC 92.1%, AUC 0.962; inter-database ACC 91.4%, AUC 0.938	Included
3	A Graph Convolutional Network Based on Univariate Neurodegeneration Biomarker for Alzheimer’s Disease Diagnosis Qu et al., 2023, IEEE JTEHM	ADNI, Aβ+ subjects; sMRI cortical thickness (FreeSurfer, left hemisphere, 34 ROIs)	GCN on a population graph (age, gender, APOE edges) with univariate neurodegeneration biomarker (UNB) node features and an attention module	Validation ACC 93.90% (AD vs CU), 82.05% (AD vs MCI)	Included
4	A hybrid deep learning framework to predict Alzheimer’s disease progression using generative adversarial networks and deep convolutional neural networks. SinhaRoy & Sen, 2024, Arab J Sci Eng	ADNI-derived 4-class set of 2D sMRI slices; 6,400 images	DCGAN (synthetic images) + SRGAN (super-resolution) + compact 2D CNN	ACC 99.7% as stated in the abstract (the paper’s Table 4 does not label its metric; the NL/AD value of 1.000 is not confirmed as AUC). Class counts match the Kaggle four-class set used in row 11	Reviewed only: not a graph method
5	An Alzheimer’s Disease classification network based on MRI utilizing diffusion maps for multi-scale feature fusion in graph convolution. Yang et al., 2024, Math Biosci Eng	ADNI; 338 AD, 422 MCI, 532 CN; sMRI ROI features (116 AAL regions)	Population graph; diffusion maps at multiple neighbour counts; metric-learned adjacency; self-attention fusion	ACC: AD-CN 0.887, AD-MCI 0.705, CN-MCI 0.825, 3-class 0.671	Included
6	Graph-based multimodal fusion framework with correlation-aware learning for Alzheimer’s disease prediction. Li et al., 2026, Big Data Min Anal	ADNI; 417 subjects (103 CN, 103 EMCI, 105 LMCI, 106 AD); sMRI + SNP genetics	Brain-region graph + gene-ontology graph; GCN with pooling; cross-attention + correlation-aware learning (CAL) module	CN vs AD: ACC about 0.82, AUC 0.82; CN vs MCI: ACC 0.773; MCI vs AD: ACC 0.747	Included
7	Longitudinal graph neural networks for prognostic modeling of Alzheimer’s disease progression. Ashimgaliyev et al., 2026, AAIML	ADNI; about 600 MCI subjects; longitudinal sMRI + FDG-PET / amyloid-PET features	Patch-based 3D CNN embeddings as nodes in a spatiotemporal graph (82 ROIs); GAT (spatial) + GRU with temporal attention	AUC 0.89, C-index 0.84, BAC 86%, SEN 85%, SPE 87%	Included
8	A multilayered framework for diagnosis and classification of Alzheimer’s disease using transfer learned Alexnet and LSTM Goyal, Rani & Singh, 2024, Neural Comput Appl	ADNI T1 MRI; 200 subjects (60 AD, 73 MCI, 67 CN); 5,980 2D slices	Pre-trained AlexNet features, then LSTM, then transfer-learned dense layers; GAN augmentation of training data	With GAN: ACC 96.83% (3-class), 98.13% (AD-CN), 99.38% (AD-MCI), 99.37% (CN-MCI)	Reviewed only: not a graph method
9	Machine Learning and Deep Learning for Alzheimer’s Disease Diagnosis: A Survey of MRI, PET, Multimodal Fusion, Transformers, Graph Neural Networks, and Explainable AI Ahmad, Tarannum & Uddin, 2026, Am J Adv Technol Eng Solut	Survey, no dataset of its own; covers ADNI, OASIS-3, AIBL	Narrative review organised by traditional ML, CNN, multimodal fusion, transformers, GNN and explainable AI	None (no experiments). Concludes multimodal usually beats single-modality; transformers and GNNs promising but not proven superior	Background: survey

#	Paper	Dataset	Method	Headline metrics	Decision
10	Graph neural networks in Alzheimer's disease diagnosis: a review of unimodal and multimodal advances Ali, Piana, Pardini & Garbarino, 2025, Front Neurosci	73 reviewed studies; mostly ADNI, also TADPOLE, OASIS, UK Biobank, ABIDE	Review organised by GNN pipeline: graph construction, convolution, pooling, prediction, explainability	None of its own. Best reported 99.3% NC/AD (TADPOLE, multimodal); multimodal studies had higher median accuracy than unimodal	Background: review of GNNs
11	Development of framework by combining CNN with KNN to detect Alzheimer's disease using MRI images Lanjewar, Parab & Shaikh, 2023, Multimed Tools Appl	Kaggle 4-class Alzheimer's MRI set; 6,400 images (ND 3,200, VMD 2,240, MID 896, MOD 64)	4-block CNN as feature extractor feeding a KNN (k = 3); compared with ResNet50, VGG16, MobileNetV2	Mean ACC 99.58% (K = 20), F1 99.43%, MCC 99.31%	Reviewed only: not a graph method